Inertia and balanced forces

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict motion when the net force is zero.

SUCCESS CRITERIA - I can...

1. Add forces to decide whether they balance.
2. Predict constant velocity when net force is zero.
3. Explain why changing direction requires net force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “net force” and “inertia” to explain your answer. Sentence starters are optional.

net force: the vector sum of all forces on one object

inertia: the tendency to keep the same velocity without net force

balanced: adding to zero net force in this force context

01 Warm-up: bring back an earlier idea

A goes **6 m** *east* and **4 m** *north*. B then goes **3 m** *west* and **8 m** *south*. **Draw the route** and **find the endpoint relative** to the start.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Sigma F = \mathbf{0} \Rightarrow a = \mathbf{0} \Rightarrow v = \text{constant}$.
ΣF means net force. Balanced forces allow rest or steady motion in a straight line.

Worked reasoning: A book has **6 N** upward support and **6 N** downward weight. Net force is zero. Its velocity stays constant; it could be resting or moving steadily.

Your turn: Draw the two force arrows. Then **sketch two possible velocity-time graphs** consistent with this balance.

Sentence starter: In my work, I used … to show … . I labeled it … .



Book: compare support and weight.

Marzano check - circle: ① ② ③

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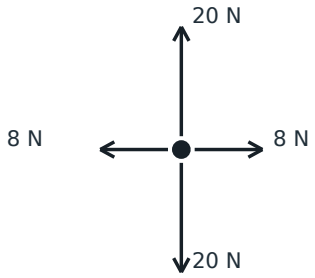
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Balanced while moving

The box is sliding right at steady velocity.
1. **Add the horizontal forces shown**.
2. Does zero net force mean the box must stop?

Sentence starter: I notice … . This supports … because … .



The dot represents the object. Box moving right

Marzano check - circle: 1 2 3

04 After release

You push a puck, then your hand loses contact. Assume a level surface with no friction or air resistance. **Complete the puck's force diagram** after release. **What happens** to the hand force and to the puck's horizontal velocity?

Sentence starter: In my work, I used … to show … . I labeled it … .

Add and label force arrows.

Puck after release; it is moving right.

Marzano check - circle: 1 2 3

05 The bus stops

A bus brakes while traveling forward.
1. **Explain why a passenger tends** to keep moving forward relative to the bus.
2. **Draw the passenger's velocity** and a separate arrow for the force from a seat belt.

Sentence starter: I think … . I can point to …, which shows …, so … .

Passenger during braking

Forward is right.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

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Continue down each column.

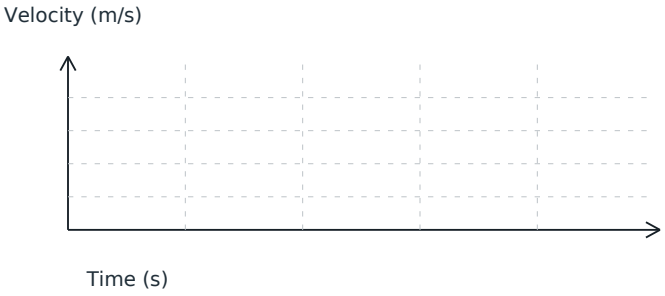
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A cart accelerates uniformly from rest to **10 m/s** *east* in **3 s**. **Find acceleration**, average velocity and displacement.
Sketch the velocity–time graph; **label the given values**.

Word help: acceleration = rate of change of velocity

Sentence starter: I found ... [unit]. This value means ... in this situation.

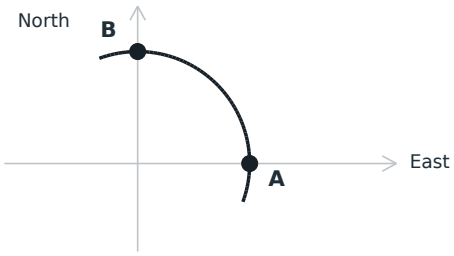


Marzano check - circle: 1 2 3

07 Exit check

Which motions can have zero net force: resting, straight at steady speed, turning at steady speed, or speeding up? For the turning case shown, draw velocity arrows at A and B and **explain your choice**.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



Path sketch: runner moves from A toward B.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A puck slides right at steady speed. **Draw one possible balanced force** diagram and **explain why a forward force** is not required in the case where friction is ignored.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



Add and label force arrows.

Puck moving right on a frictionless level surface.

Marzano check - circle: 1 2 3